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Regular Research Manuscript

A Solar PV System for Industrial Application- A Case Study of a Cement Industry in TANZANIA

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ABSTRACT

This study presents an implementation of the largest behind-the-meter solar photovoltaic (PV) system for a cement industry in Tanzania. The project aims to supply energy for the industry's own use, without any energy injection into the utility grid, for the purposes of cutting energy-related expenditure and reducing reliance on the utility grid. The project also aims to contribute to the efforts towards the application of renewable and alternative energy sources as a means to reducing carbon footprints, towards achieving the net-zero CO_2 emission goal. A comprehensive site assessment was conducted, analyzing factors such as location, land availability, climate data, and environmental considerations. The solar resource assessment involved analyzing Global Horizontal Irradiation (GHI) data and evaluating inter-annual variability and uncertainty. The energy yield assessment, utilizing PVsyst software, estimated annual energy production and performance ratios. The plant design outlines the PV plant configuration, including solar modules, inverters, and transformers. The findings of the study demonstrate the technical viability of a 15 MWp solar PV system for the cement industry, showcasing its potential to reduce reliance on grid electricity and promote sustainable energy practices.

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Highlights:

- The paper presents the implementation of a behind-the-meter solar PV system for the cement industry in Tanzania.
- A case of a 15 MWp grid-tied PV plant designed to cater for the cement plant's own consumption without grid export is presented.
- Key performance indicators, including annual energy yield, performance ratio, and system losses, are evaluated and discussed.
- Results confirm the potential of industrial solar PV systems to reduce grid dependence, operational energy costs, and carbon footprints.

INTRODUCTION

In recent times, the electrical power and the global energy sector in general is witnessing a major shift in consumption trends towards renewable energy resources (REN21, 2024) . This results from growing concerns about the increasing energy demand, costs, and depletion of fossil fuels, which for many years have been the conventional energy sources (Dirie et al., 2025). The shift is also largely contributed to by the rising concerns of global climate change caused by emissions of greenhouse gases (GHG) highly contributed by the use of fossil fuels (Gawhade and Ojha, 2021a). The Industrial sector is among some of the most carbon-intensive industries, which make up a significant part of the economy and a large contribution to energy consumption, resource consumption, and greenhouse gas emissions (Andersson, 2020).

The year 2024 witnessed the largest ever recorded growth of global energy demand by 2.2% along with an increase in energy-related CO_2 emissions of 0.8%, hitting a 37.8 $GtCO_2$ emission. The electricity demand grew by 4.3%, a more rapid growth than the overall energy demand increase. Renewables accounted for the largest share of the growth in total energy supply (38%), which was followed by natural gas (28%) (IRENA, 2024)).

Nearly all the rise in electricity demand in 2024 was met by renewable energy sources, led by solar PV, which witnessed the highest capacity expansion (IEA, 2025). This reflects a growth in renewable energy technologies and their acceptance as a solution to the rising energy demand and the global climate change problem. The growth of the renewable energy share in electricity production has also contributed to the decreased costs and bureaucracy in the implementation of renewable energy projects (Minja and Mushi, 2023). For example, for the year 2023, the global average cost of electricity (LCOE) from solar PV fell by 12% (IRENA, 2024)

In this study, a grid-tied solar PV plant is explored as a solution to the energy crisis and the global climate change problem. It involves the feasibility study and design of a grid-tied solar PV plant for industrial consumption using PVsyst software, with consideration of reducing the energy-related GHG emissions and savings on the energy cost expenditure. A case study of a cement factory is considered where a 15MWp grid-tied solar PV plant is installed.

Solar PV Systems

Solar PV accounts for one of the fastest-growing renewable energy technologies. By the end of the year 2023, about 1,411GW of solar PV systems had been installed all over the World. The rapid growth is largely contributed by the continued improvement in technology, and design innovation leading to improved competitiveness and a significant reduction in solar PV installation costs. The global average cost of electricity from utility-scale solar PV fell to USD 0.044 per kilowatt-hour (kWh) in the year 2023 (IRENA, 2024).

Rapid advancement is also contributed by the fact that solar energy is one of the most abundant renewable energy resources in most parts of the world, which is relatively easy to harness. In many countries, including Tanzania, the potential for solar PV electricity generation greatly surpasses the country's electricity demand (ESMAP, 2020).

Grid-tied solar PV systems offer one of the best ways to harness the abundant solar energy while staying connected to the utility grid. They allow users to generate and consume electricity from their solar plants, while feeding any excess power to the grid, thereby improving the system's energy mix and easing the burden on the sources that supply power to the grid. Grid-tied solar PV systems are also claimed to be more cost-effective and require less maintenance as compared to Battery Energy Storage (BESS) systems (Streedevi et al., 2016).

METHODS AND MATERIALS

Case Study Description

A 15 MWp solar PV plant is installed for a cement plant in Tanzania. The plant is erected on a site covering approximately 19 hectares of land and located 3km away from the cement factory. The designated area houses facilities for power generation, control, and monitoring. Figure 1 shows the location of the PV plant erection. The cement plant is connected to the utility grid via a 132 kV line and operates a 132/3.3 kV electrical substation, shown in Figure 2. The installed solar PV is connected to this substation through a constructed 3 km medium voltage (MV) overhead line within the cement quarry.

Figure 1: Site location.

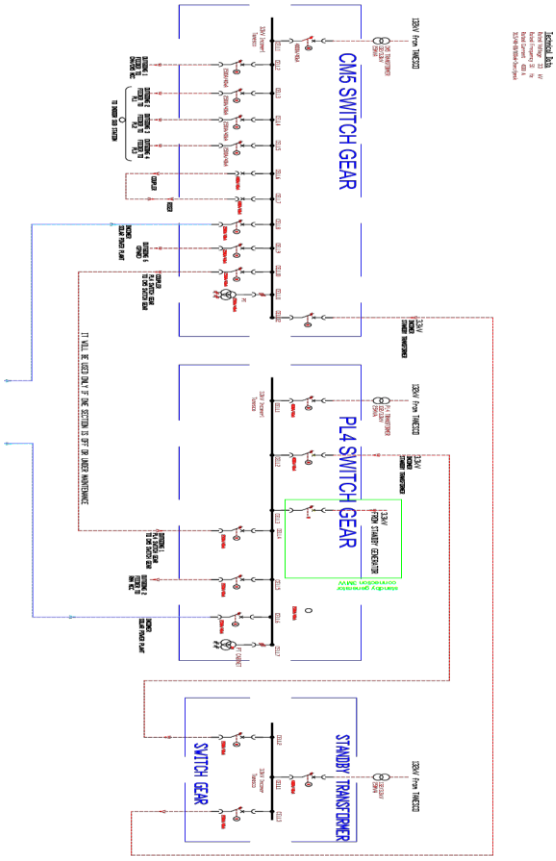

Figure 2: Existing substation single line diagram.

The line extends from the photovoltaic systems' low-medium voltage (LV/MV) step-up transformers to an MV/LV transformation cabin. A 3.3 kV connection is established between the two substations using the previously existing 132/3.3 kV substation. The purpose of the photovoltaic plant is to generate energy only for the own use of the cement factory, without any energy injection in the TANESCO grid: the size of the PV plant is calibrated in order to serve the factory loads without producing excess energy.

Site Solar Potential

Climatic Data

The project's region experiences a tropical climate with distinct wet and dry seasons. Average temperatures range from 23 °C to 32 °C throughout the year. The city's climate is influenced by monsoon winds from the Indian Ocean, and sea breezes help moderate temperatures. Climatic data related to the performance of energy production at the site, as measured by a NASA satellite, includes a mean annual ambient temperature of about 26°C. Table 1 presents climatic data for the site for the whole year.

Table 1: Site's climatic data

Month	GHI (kWh/m²/day)	DHI (kWh/m²/day)	Ambient Temperature (°C)
Jan	5.76	2.17	26.8
Feb	6	2.19	26.9
Mar	5.34	2.25	26.8
Apr	4.57	2.08	26.6
May	4.42	1.83	26.3
Jun	4.56	1.66	25.7
Jul	4.55	1.73	25
Aug	4.86	1.92	24.8
Sep	5.57	2.07	25
Oct	5.67	2.23	25.5
Nov	5.7	2.19	26
Dec	5.71	2.14	26.5

Solar Resource Assessment

A comprehensive solar resource assessment was conducted to evaluate the solar energy potential at the site. Global Horizontal Irradiation (GHI) data from multiple sources, including Meteonorm, NASA-SSE, and PVGIS, were analyzed as presented in Table 2 in kWh/m2/day. The inter-annual variability of solar resources was determined to be 2.5%, indicating a consistent solar resource.

Table 2: Irradiation data

Month	Meteonorm 8.1	NASA-SSE	PVGIS TMY 5.2
Jan	6.26	5.76	7.05
Feb	6.42	6	7.10
Mar	6	5.34	6.13
Apr	5.45	4.57	3.94
May	4.82	4.42	4.44
Jun	5.02	4.56	4.54
Jul	5.46	4.55	5.03
Aug	6.01	4.86	5.48
Sep	6.32	5.57	6.61
Oct	6.6	5.67	6.98
Nove	6.45	5.7	6.48
Dec	5.81	5.71	5.45

PV Plant Design

A solar power plant consists of different components, including solar modules on mounting structures, inverters, step-up transformers, and overhead line connection interfaces. Solar modules convert solar radiation into electricity. A power plant consists of several modules connected in strings to produce the required DC power output. Strings are then connected to central inverters that convert DC power into alternating current (0.8kVAC) for connection to step-up transformers, where it will be stepped up to 33kV.
Electricity is then fed into an overhead line. The overhead line connection interface, at the substation, is where electricity will be exported into the factory. The substation will have the required grid interface switchgear, such as circuit breakers (CBs) and disconnectors for protection and isolation of the PV power plant, as well as metering equipment. Figure 3 shows a pictorial presentation of a proposed solar plant. The technical specifications for the PV plant are summarized in Table 3.

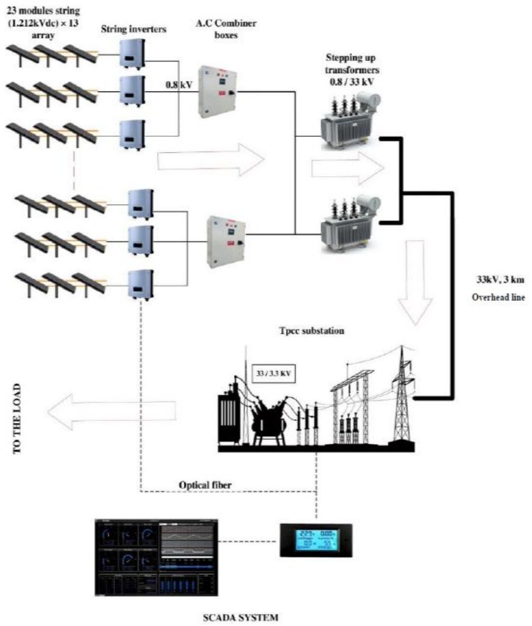

Figure 3: PV plant pictorial representation.

Solar PV: To generate 15 MWp plant capacity, the plant consists of 615 Wp solar PV modules (JINKO JKM615M-7RL4-V) arranged into strings of 28 modules connected in series. A total of 4 arrays is produced with the output nominal power of 15.016

MWp (STC). The nominal DC power is estimated according to *Equation 1* (IEC, 2017).

Table 3: Plant components parameters

Parameter	Value
Plant Capacity	15 MWp
Solar Modules	N-type, Bifacial, 615 Wp
Number of Modules	24416
Solar Mounting Technology	Fixed tables
Inverter Technology	320 kWac, string inverter
Number of inverters	40

$$P_{nom} = N_{arr} \times N_{\mathrm{mod}} \times P_{\mathrm{mod}} \quad (1)$$

where P_{nom} represents the nominal power of the PV array (W), N_{arr} and N_{mod} the number of parallel module strings (arrays) and modules per array, respectively, and P_{mod} is the rated power of one PV module (Wp).

The solar modules used for the project are N-type, which can have an efficiency level up to 25.7%, as compared to the 23.6% efficiency of the P-type panels. The modules are bifacial, which enables them to utilize the albedo effect, where light is reflected off the ground or other surfaces (Jinko Solar, 2022).

Cabling: Cables are selected by considering the extent of voltage drop for a particular cross-sectional size of the cable. Ensuring that the Voltage drop of cables in the PV plant is less than 1% (Bhatia, 2014). The allowable voltage drop is obtained from the maximum current at STC (I_{mpp}), cable's resistance per meter (R), cable length (L), and the input voltage (V_{in}) as in *Equation 2*.

$$\Delta V(\%) = \frac{2I_{mpp}RL}{V_{in}} \times 100 \quad (2)$$

where $\Delta V(\%)$ is the percentage voltage drop, I_{mpp} the current at the maximum power point at STC and DC input voltage V_{in} while L and R are the one-way cable length in meters and the cable resistance per unit length, respectively. Factor 2 accounts for the forward and return conductors.

Inverters: A total of 40 string inverters (HUAWEI SUN2000-330KTL-H1) are used for the conversion of DC to AC power at 0.8 kV, 10 for each PV array. The total number of inverters is obtained as in *Equation 3*.

$$N_{inv} = \frac{N_{str}}{N_{str,inv}} \quad (3)$$

with N_{inv} being the number of inverters per array, while N_{str} is the total number of strings, and $N_{\mathrm{str,\,inv}}$ is the number of strings per inverter.

Each inverter has an operating voltage range of 500-1500 V. The AC output power of the inverters is given by *Equation* **4.**

$$P_{AC} = \frac{P_{DC}}{DC\,/\,AC\ ratio} \quad (4)$$

P_{AC} being the inverter AC output power (W), P_{DC} The DC input power to the inverter (W) with DC/AC ratio as the inverter loading ratio.

The AC output determines the size of the inverter to be used, and in this case, from the calculations, an inverter's nominal power of 300 kWac is chosen.

Transformers: Four 6.6 MVA step-up transformers are used to increase the AC voltage to 33 kV for efficient transmission. The transformer's output is connected to a common bus bar for power evacuation from the field to the factory substation. At the factory end, two 15 MVA step-down transformers reduce the voltage to 3.3 kV and feed the factory feeders for synchronization.

Transmission line: A single circuit 3 km, 33 kV overhead line connects the solar plant's substation to the cement factory's substation.

Substations: The substation at the cement factory features switchgear, including circuit breakers and disconnectors, for the protection and isolation of the PV power plant, as well as metering equipment. The PV plant's single-line diagram is shown in Figure 4.

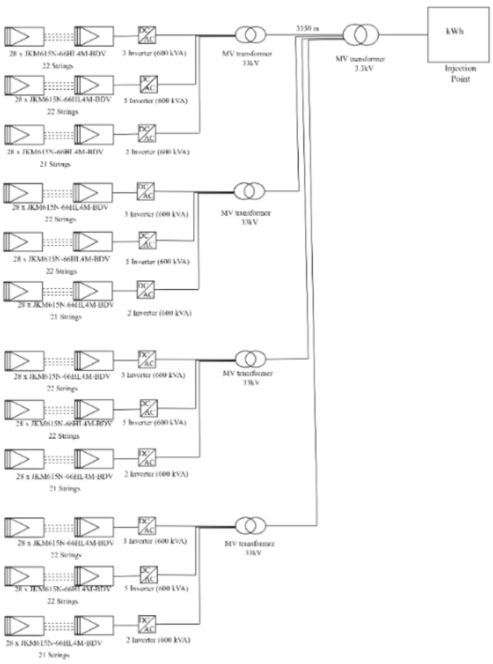

Figure 4: Solar plant single line diagram.

RESULTS AND DISCUSSIONS

Annual Energy Production

The energy yield assessment of the plant was conducted using the global standard software for assessing solar PV performance, PVsyst. The system's output power distribution curve is shown in Figure 5. From which it is estimated that the plant will produce 12.7 MW, with the Annual production probability shown in Figure 6.

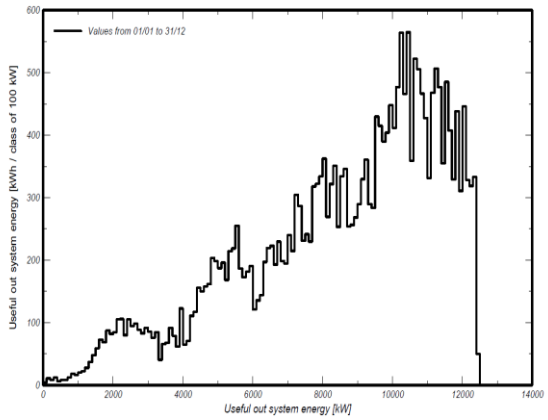

Figure 5: System's output power distribution curve.

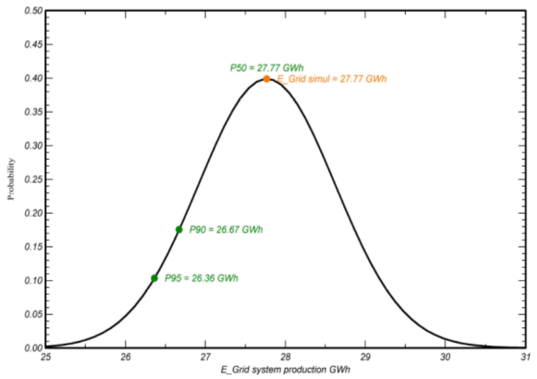

Figure 6: Plant's annual production probability.

From the annual production probability curve, the P50, P90, and P95 values were estimated to be at 27.77 GWh/y, 26.67 GWh/y, and 26.36 GWh/y, respectively, with a variability of 0.83 GWh. The P50, P90, and P95 values represent the energy production levels expected with 50% 90%, and 95% certainty, respectively.

Losses

The system's loss diagram from the PVsyst simulation is shown in Figure 7. Total system losses accounted for are optical losses, electrical losses in the PV, wiring, and transformers, and environmental and design losses due to unavailability and degradation losses.

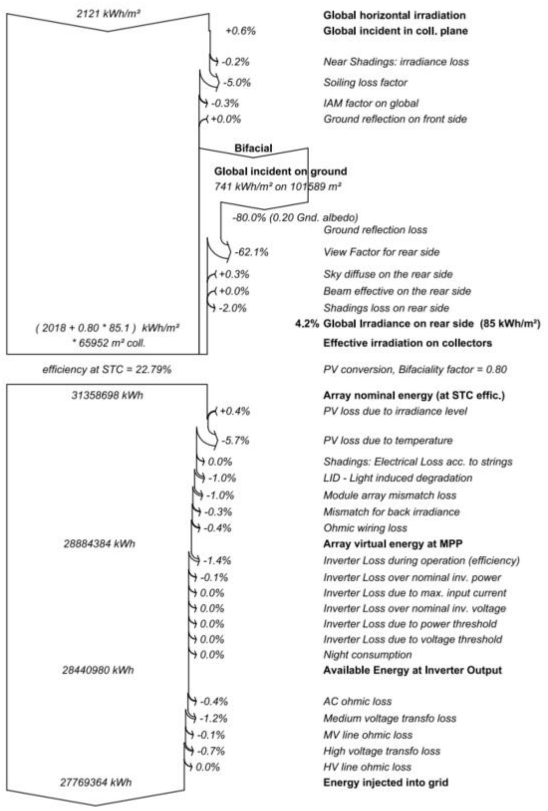

Figure 7: Loss diagram.

The total irradiation on the solar panels is contributed by both the incident and reflected irradiation at 0.2 albedo. With the efficiency of about 22.7% at STC, about 31.36 GWh converted by the solar plant, some of which is lost in the PV itself, the inverter, wiring, transformers, and transmission lines, leaving approximately 27.77GWh/y that is injected into the plant's distribution. The array losses assumptions included an array soiling loss of 5.0%, light-induced degradation loss fraction of 1.0% and module mismatch loss fraction of 1.0% at MPP.

Performance Ratio

The Performance Ratio (PR) measures the quality of a solar installation, reflecting how well the system converts sunlight into usable electrical energy. It measures the relationship between the actual energy output of a solar installation and its theoretical maximum potential output under Standard Test Conditions (STC) of solar irradiance of 1000 watts per square meter, an air mass of 1.5, and a module temperature of 25°C. In the simulation made with PVsyst, the following PR values are provided for an annual mean of 84.7%. The normalized monthly PR values are shown in Figure 8.

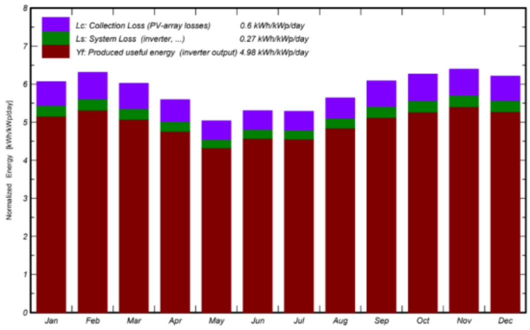

Figure 8: PR values for the solar plant.

The PR graphs seen in Figure 8 above exhibit seasonal variability due to changes in solar irradiance, ambient temperature, and environmental conditions throughout the year. Studies have shown that PR fluctuates with season because of meteorological variations such as differences in irradiance intensity, angle of the sun, and temperature, which influence module efficiency and system losses (Phinikarides et al., 2015). For instance, long-term monitoring of grid-connected PV systems revealed distinct seasonal patterns in PR, with higher values occurring in seasons with milder temperatures and low values of extreme temperature periods due to increased thermal losses and spectral shifts in solar radiation (Elhadj Sidi et al., 2016).

In real environments, such as the quarry to which the PV plant is installed, it is anticipated that the environmental factors, such as dust accumulation, shading, and soiling, will further contribute to PR variations. Dust decomposition on PV module surfaces can significantly reduce current and voltage output, with reported power reduction of nearly 8-12% depending on dust levels and local conditions (Mustafa et al., 2020).

To block the wind and reduce dust levels, over 1,500 Murraya tree species will be planted on the sides around the solar plant.

The performance of this plant is useful to compare the key performance indicators, such as the PR and specific yield, with data from similar PV installations in the tropical and semi-arid regions. For a grid-connected solar PV system operating in Saharan and hot climates, the performance ratio ranged from approximately 66% to 84.5% depending on the location and system design (Hazem et al., 2025). This suggests that the PR within this range is characteristic of high solar irradiance in the regions, supporting the comparability of the Cement plant's PR results with the real-world benchmarks.

The PR of Cement PV systems fall above the commonly observed ranges of PV plants in tropical to semi-arid climates, suggesting an effective design. These comparisons also underscore the influence of site-specific factors such as module technology, array configuration, and maintenance practices on measure performance.

CONCLUSION

The study demonstrates the technical viability of a 15 MWp solar PV system producing an estimated 12.7 MW of power for a cement industry in Tanzania. The PVsyst simulation provided key energy yield metrics, including P50, P90, and P95 production values and the Performance Ratio, which indicate the system's efficiency with an estimated 27.77 GWh annual energy production. The project has the potential to provide a sustainable and cost-effective energy solution, reducing the industry's reliance on grid electricity and promoting the adoption of renewable energy. The successful implementation of this project can serve as a model for other industries in Tanzania and contribute to the country's decarbonization efforts.

For future work, it is recommended that cement and other energy-intensive industries expand renewable energy integration through larger PV capacities or hybrid systems coupled with energy storage to enable time-shifting, peak load management, and enhanced reliability. Complementary energy efficiency measures, such as process optimization and waste heat recovery, should be implemented to reduce overall energy demand. Real-time monitoring and predictive analytics can optimize system performance, while staff training ensures proper operation and maintenance. Additionally, developing long-term decarbonization roadmaps, engaging with supportive policies and financing mechanisms, and incorporating renewable energy metrics into sustainability reporting will further accelerate a clean and sustainable energy transition for industrial facilities.

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